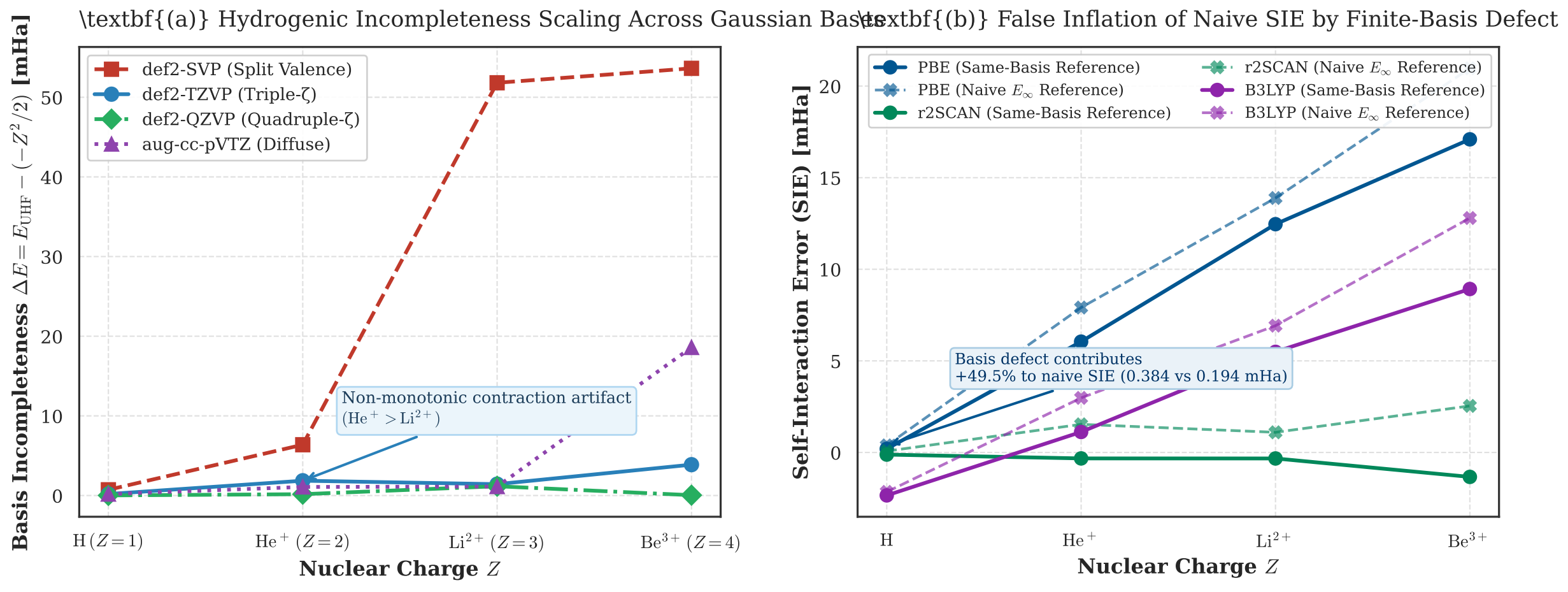
\textbf{(a)} Hydrogenic Incompleteness Scaling Across Gaussian Basis Sets
Basis Incompleteness $\Delta E = E_{\mathrm{UHF}} - (-Z^2/2)$ [mHa]
def2-SVP (Split Valence)
def2-TZVP (Triple-ζ)
def2-QZVP (Quadruple-ζ)
aug-cc-pVTZ (Diffuse)
Non-monotonic contraction artifact
($He^+ > Li^{2+}$)
0
10
20
30
40
50
H ($Z=1$)
He^+ ($Z=2$)
Li^{2+} ($Z=3$)
Be^{3+} ($Z=4$)
Nuclear Charge Z
\textbf{(b)} False Inflation of Naive SIE by Finite-Basis Defect
Self-Interaction Error (SIE) [mHa]
PBE (Same-Basis Reference)
PBE (Naive E_∞ Reference)
r2SCAN (Same-Basis Reference)
r2SCAN (Naive E_∞ Reference)
B3LYP (Same-Basis Reference)
B3LYP (Naive E_∞ Reference)
Basis defect contributes
+49.5% to naive SIE (0.384 vs 0.194 mHa)
0
5
10
15
20
H
He^+
Li^{2+}
Be^{3+}
Nuclear Charge Z